

Lab 3: Relational Algebra

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1. Relational Algebra Queries

Query 1

$$\begin{aligned} & \rho(Makers, \pi_{madeByPid}(Reservation)) \\ & \pi_{firstName, lastName}(Makers \\ & \quad \bowtie_{Makers.madeByPid=Person.pid} Person) \end{aligned}$$

Query 2

$$\begin{aligned} & \rho(InLab, \pi_{pid}(Attached)) \\ & \rho(PIs, \pi_{pid}(\sigma_{role='PI'}(Participates))) \\ & InLab \cup PIs \end{aligned}$$

Query 3

$$\begin{aligned} & \rho(CPULabs, \pi_{labId}(\sigma_{category='CPU Server'}(EquipmentModel) \\ & \quad \bowtie_{EquipmentModel.modelId=EquipmentUnit.modelId} EquipmentUnit)) \\ & \rho(CoCLabs, \pi_{labId}(\sigma_{building='College of Computing'}(Laboratory))) \\ & CPULabs \cup CoCLabs \end{aligned}$$

Query 4

$$\begin{aligned} & \rho(CPULabs, \pi_{labId}(\sigma_{category='CPU Server'}(EquipmentModel) \\ & \quad \bowtie_{EquipmentModel.modelId=EquipmentUnit.modelId} EquipmentUnit)) \\ & \rho(GPULabs, \pi_{labId}(\sigma_{category='GPU Server'}(EquipmentModel) \\ & \quad \bowtie_{EquipmentModel.modelId=EquipmentUnit.modelId} EquipmentUnit)) \\ & CPULabs \cap GPULabs \end{aligned}$$

Query 5

$$\begin{aligned} & \rho(A, \pi_{supervisorPid, labId}(Laboratory)) \\ & \rho(B, \pi_{labId}(\sigma_{building='College of Computing'}(Laboratory))) \\ & Faculty \bowtie_{Faculty.pid=supervisorPid} (A/B) \end{aligned}$$

Query 6

$$\begin{aligned}
& \rho(A, \pi_{pid, projectCode}(Participates)) \\
& \rho(LabBudgets, \pi_{budgetLine}(\sigma_{labId=2}(FundsLab))) \\
& \rho(B, \pi_{projectCode}(LabBudgets \\
& \quad \bowtie_{LabBudgets.budgetLine=FundsPrj.budgetLine} FundsPrj)) \\
& A/B
\end{aligned}$$

Query 7

$$\begin{aligned}
& \rho(C, \gamma_{madeByPid}(COUNT(resId) \rightarrow n)(Reservation)) \\
& \rho(C_1(1 \rightarrow p_1, 2 \rightarrow n_1), C) \\
& \rho(C_2(1 \rightarrow p_2, 2 \rightarrow n_2), C) \\
& \pi_{p_1, p_2}(C_1 \bowtie_{n_1 > n_2} C_2)
\end{aligned}$$

Query 8

$$\begin{aligned}
& \rho(R, \pi_{madeByPid, labId}(Reservation \\
& \quad \bowtie_{Reservation.resId=Reserves.resId} Reserves \\
& \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit)) \\
& \rho(G, \gamma_{madeByPid}(COUNT(labId) \rightarrow k)(R)) \\
& \pi_{madeByPid}(\sigma_{k \geq 2}(G))
\end{aligned}$$

*Assumption: the laboratory of a unit is its current location (**EquipmentUnit.labId**), since the schema keeps no location history.*

Query 9

$$\begin{aligned}
& \rho(Sep, \sigma_{submissionTS \geq '2026-09-01' \wedge submissionTS < '2026-10-01'}(Reservation)) \\
& \rho(AI, \sigma_{name='AI Movement'}(Laboratory)) \\
& \pi_{Sep.resId}(Sep \\
& \quad \bowtie_{Sep.resId=Reserves.resId} Reserves \\
& \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit \\
& \quad \bowtie_{EquipmentUnit.labId=AI.labId} AI)
\end{aligned}$$

Query 10

$$\begin{aligned}
& \rho(P, \pi_{pid, projectCode}(\sigma_{role='PI'}(Participates))) \\
& \rho(G, \gamma_{pid}(COUNT(projectCode) \rightarrow n)(P)) \\
& \pi_{pid}(\sigma_{n > 1}(G))
\end{aligned}$$

Query 11

$$\begin{aligned}
& \rho(R, \pi_{approverPid, labId}(\sigma_{status='Approved' \vee status='Completed'}(Reservation) \\
& \quad \bowtie_{Reservation.resId=Reserves.resId} Reserves \\
& \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit)) \\
& \rho(G, \gamma_{approverPid}(COUNT(labId) \rightarrow k)(R)) \\
& \pi_{approverPid}(\sigma_{k > 1}(G))
\end{aligned}$$

Assumption: a reservation counts as approved if its status is Approved or Completed.

Query 12

$$\begin{aligned} & \rho(Busy(1 \rightarrow pid), \pi_{madeByPid}(\sigma_{plannedStart \leq '2026-11-06\ 23:59:59' \wedge plannedEnd \geq '2026-11-06\ 00:00:00'}(Reservation))) \\ & \pi_{pid}(Person) - Busy \end{aligned}$$

Query 13

$$\begin{aligned} & \rho(C, \gamma_{labId}(COUNT(serialNo) \rightarrow n)(EquipmentUnit)) \\ & \rho(Avg, \gamma(AVG(n) \rightarrow a)(C)) \\ & \rho(Low, \pi_{labId}(\sigma_{n < a}(C \times Avg))) \\ & Laboratory \bowtie_{Laboratory.labId=Low.labId} Low \end{aligned}$$

Query 14

$$\begin{aligned} & \rho(R, \pi_{labId, madeByPid, resId}(\sigma_{status='Approved' \vee status='Completed'}(Reservation) \\ & \quad \bowtie_{Reservation.resId=Reserves.resId} Reserves \\ & \quad \bowtie_{Reserves.serialNo=EquipmentUnit.serialNo} EquipmentUnit)) \\ & \rho(C, \gamma_{labId, madeByPid}(COUNT(resId) \rightarrow n)(R)) \\ & \rho(M(1 \rightarrow labM, 2 \rightarrow m), \gamma_{labId}(MAX(n) \rightarrow m)(C)) \\ & \rho(Best, \pi_{labId, madeByPid}(C \\ & \quad \bowtie_{C.labId=M.labM \wedge C.n=M.m} M)) \\ & Best \bowtie_{Best.madeByPid=Person.pid} Person \end{aligned}$$

Assumption: ties are kept, so a laboratory may return several persons.

Query 15

$$\begin{aligned} & \rho(R, \sigma_{submissionTS \geq '2025-01-01' \wedge submissionTS < '2026-01-01'}(Reservation)) \\ & \rho(G, \gamma_{madeByPid}(COUNT(resId) \rightarrow n)(R)) \\ & \pi_{madeByPid}(\sigma_{n \geq 3}(G)) \end{aligned}$$

2. Challenges

Getting started

At first I did not know how to turn a sentence into an algebra expression, and I also had to learn how to write it in $\text{\LaTeX}$. Each query took a long time, because I kept going back to the Lab 2 solution to check the exact names of tables and attributes, and sometimes to Lab 1 to understand the assumptions. A small mistake like `madeBy` instead of `madeByPid` makes the query wrong. I started testing each query on a few example rows, and this helped me find my mistakes.

Joins

I first wanted to use a natural join, but `madeByPid` and `pid` have different names, so nothing matches. I wrote every join condition explicitly. For Query 7 I needed the same relation twice, so I used ρ to rename the columns.

Union, intersection, and, or

Queries 3 and 4 looked the same to me, but they are not. In Query 4 I first used `CPU Server` $\vee$ `GPU Server`, and it returned laboratories with only one of the two. With $\wedge$ it returns nothing, because a unit has only one category. I understood that σ checks one row at a time, so I built one set of laboratories for CPU and one for GPU, and used $\cap$.

Every and no

For “every” (Queries 5 and 6) I used division. The difficult part was choosing A and B . For Query 12 I did not understand at first why we subtract: σ cannot find persons who have no matching row, so I computed the persons who have a reservation on November 6 and removed them from all persons.

Counting and average

Aggregation was new for me. σ and π look at one row, and γ works on groups of rows. For “at least three” I use γ first, then σ . In Query 13 I asked why I could not just write the average value. The reason is that the average has to be computed inside the query, and since σ compares columns of the same row, I attach it to every laboratory with a cross product.

Choices about the data

- A unit only stores its current laboratory, so Queries 8, 9, 11 and 14 use the current location.
- The reservation table contains all statuses. For approval (Queries 11 and 14) I count Approved and Completed.
- γ does not create groups for persons or laboratories with zero rows (Queries 7 and 13).